

HEALTHCONNECT CLINIC

Final Analytics & Decision Support

DATA ANALYTICS TRACK: WEEK 8 FINAL PRESENTATION

What booking behaviour tells us about who misses their appointment, and what to do about it.

The Problem

HealthConnect Clinic loses nearly half its scheduled appointments to no-shows and cancellations, and until now, it hasn't known which bookings are most at risk.

How can HealthConnect use data to reduce missed appointments and improve the patient support experience?



53.7%

Lost Slot Rate : appointments that never resulted in a completed visit



48.5%

Overall No-Show Rate across 5,000 tracked appointments

Data & Method



The Dataset

5,000 HealthConnect appointments, booking lead time, reminder channel, appointment type, distance to clinic, prior no-show history.



The Tools

Power Query for data preparation and calculated columns; Power BI and DAX for KPIs and the interactive dashboard.



The Approach

Every KPI and segment was independently validated against the raw data before being carried into this presentation.

Final KPIs

Every figure below reconciles exactly against the raw appointment data.



48.5%

Overall No-Show Rate



60.5%

30+ Day No-Show Rate



55.4%

Repeat No-Show Rate

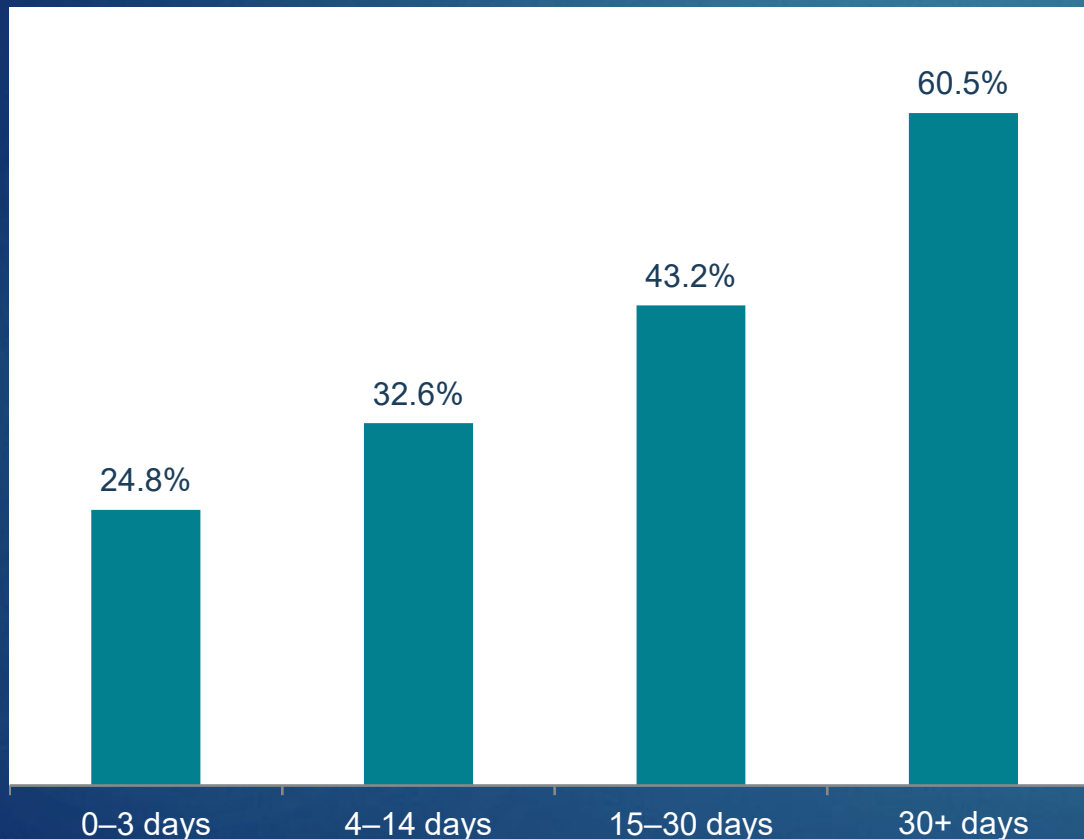


53.7%

Lost Slot Rate

The Strongest Pattern: Booking Lead Time

No-show rate more than doubles between a next-week booking and one made a month or more out.



Why it matters

This progression held up across every slicer combination I tested during Week 7, it's a genuine behavioural pattern, not a data artifact. It's also the single strongest lever in the dataset, ahead of reminder channel, appointment type, or distance to clinic.

Compounding Risk: Prior No-Show History

Within the highest-risk lead-time band, a patient's own history sharpens the picture further.

67.9%

Has prior no-show

1,015 appointments

55.2%

No prior no-show

1,410 appointments



48.5%

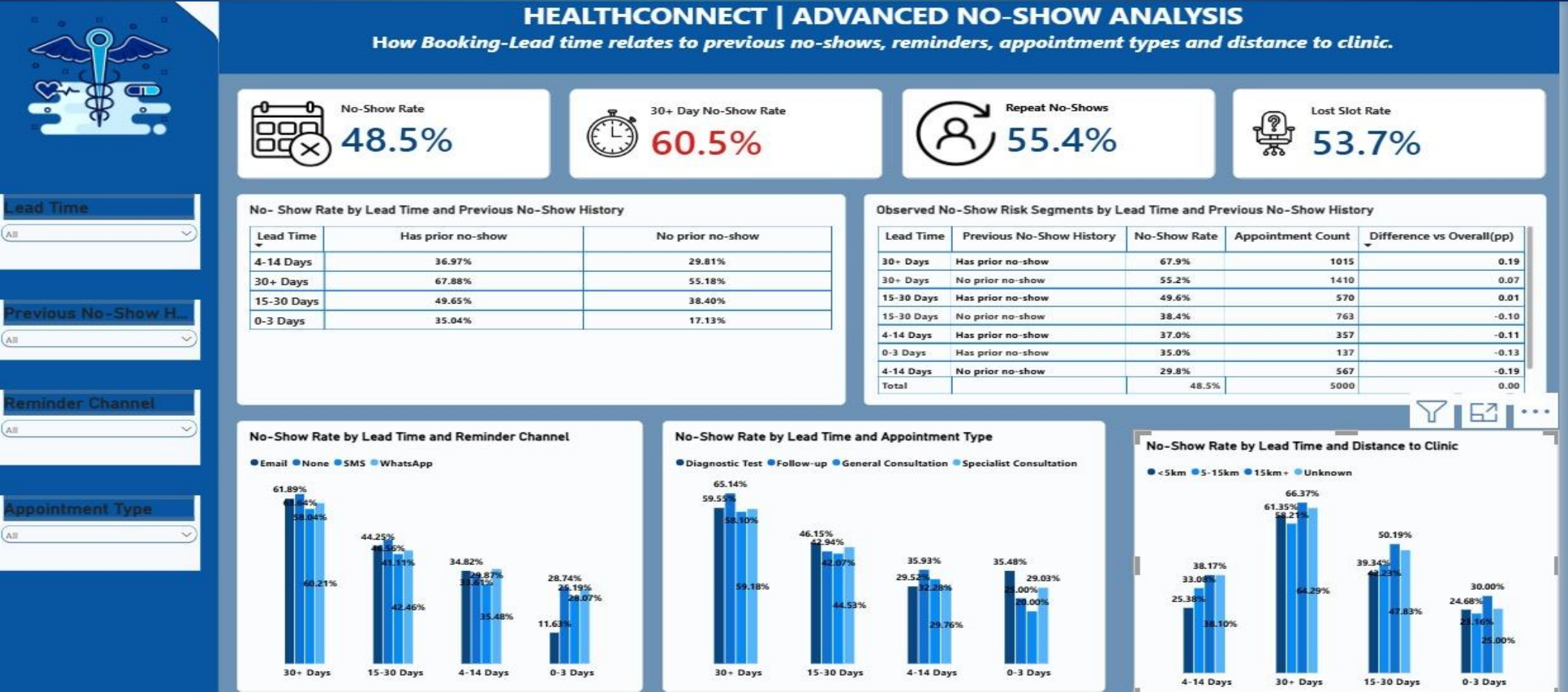
Overall baseline for comparison

30+ day bookings, split by history

Both 30+ day groups sit well above the clinic-wide baseline, long lead time is a risk factor on its own, and prior history compounds it rather than replacing it.

The Final Dashboard

Refined and re-tested in Week 7 : interactive slicers, corrected KPI logic, and a corrected distance-to-clinic sort order.



Testing & Refinement

Two real issues were found during Week 7 testing, both diagnosed, fixed, and re-tested.



Repeat No-Show KPI slicer bug

Before: Stuck at 55.4% regardless of the “No prior no-show” filter selection

After: Refined DAX — now correctly returns 0.0% for that selection



Distance-to-clinic sort order

Before: Chart legend displayed out of logical distance order

After: Custom sort column applied in Power Query : order corrected

Recommendations



Add an early touchpoint for 30+ day bookings

A confirmation reminder earlier in the lead time, not just the standard single reminder for any appointment booked more than 30 days out.



Target the highest-risk segment directly

1,015 appointments (30+ days, prior no-show) carry a 67.9% no-show rate. A phone confirmation or shorter re-booking window would concentrate effort where it matters most.

Even a modest reduction within these concentrated high-risk segments would free up more capacity than an across-the-board reminder change — because the risk isn't evenly spread.

Cross-Track Collaboration



Data Analytics

Shared the Lead Time × Previous No-Show History finding: 67.9% vs. 55.2% within the 30+ day segment, validated and re-confirmed in Week 7 testing.



Data Science

Identified this as a candidate interaction feature for their predictive model, on the basis that the two variables appear to interact rather than act independently.



Open dependency: No model-performance response received yet. Documented as an open item, not a completed collaboration, to be closed out before the HC-POD walkthrough.

Limitations



Dataset is fictional and synthetic — for learning purposes, not clinical decision-making.



Findings describe observed associations, not proven causation.



Distance and waiting-time fields contain some missing values, treated cautiously.



Reminder-channel differences are descriptive, not evidence that one channel causes better attendance.



**Testing doesn't just confirm your work is right ,
it actively looks for where it might be wrong.**

Two genuine issues were found and fixed this way, issues that would have shipped silently otherwise.

Thank you